Zeyu (Steven) Zhang

Ph.D. Student in Statistics and Data Science, Northwestern University

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Research Interests

Trustworthy and reliable large language models. My current research centers on **temporal knowledge leakage in LLM backtesting**: detecting and attributing contamination in model reasoning with interpretable, claim-level metrics, and eliminating it through inference-time supervision and reinforcement-learning post-training. Related interests include black-box LLM interpretability and auditing, RL-based fine-tuning (GRPO, DPO, RLHF), and factual memorization in LLMs.

Education

Northwestern University — **Evanston, IL, USA**

Ph.D. in Statistics and Data Science *GPA: 3.95/4.00* — *Sept 2023 – Present*

- Advisor: Prof. Bradly C. Stadie. Expected graduation: June 2028.

University of Science and Technology of China (USTC) — **Hefei, China**

B.Eng. in Electronic Information Engineering; Minor in Artificial Intelligence *GPA: 3.93/4.30* — *Sept 2019 – Jun 2023*

- Rank 5/213 in the School of Information Science and Technology.
- China National Scholarship, Ministry of Education of the PRC (top 1%).
- Wang Xiaomo Talent Program in Cyber Science and Technology; Talent Program in Artificial Intelligence.

Publications & Preprints

Conference Proceedings

[C1] **LAMP: Extracting Locally Linear Decision Surfaces from LLM World Models**
Ryan Chen, Youngmin Ko, **Zeyu Zhang**, Catherine Cho, Sunny Chung, Mauro Giuffré, Dennis L. Shung, Bradly C. Stadie
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026. ***Spotlight*** [arXiv] [OpenReview]

[C2] **Unified Off-Policy Learning to Rank: a Reinforcement Learning Perspective**
Zeyu Zhang, Yi Su, Hui Yuan, Yiran Wu, Rishab Balasubramanian, Qingyun Wu, Huazheng Wang, Mengdi Wang
Advances in Neural Information Processing Systems (NeurIPS), 2023. [arXiv] [OpenReview] [Code]

Preprints (Under Review)

[P1] **Temporal Leakage in LLM Backtesting: From Statistical Modeling to Leakage-Adjusted Evaluation**
Zeyu Zhang, Bradly C. Stadie
In submission to TMLR.

[P2] **TEMPO: Temporal Enforcement via Mode-Separated Policy Optimization for Trustworthy LLM Backtesting**
Zeyu Zhang, Bradly C. Stadie
Under review at NeurIPS 2026. [arXiv:2605.18843] [Code]

[P3] **All Leaks Count, Some Count More: Interpretable Temporal Contamination Detection and Mitigation in LLM Backtesting**
Zeyu Zhang, Ryan Chen, Bradly C. Stadie
Under review at EMNLP 2026. [arXiv:2602.17234] [Code]

Research Experience

Trustworthy LLM Backtesting: Leakage Theory, Detection & Enforcement — **Northwestern University**

Advisor: Prof. Bradly C. Stadie *→ papers [P1], [P2], [P3]* — *Sept 2025 – Present*

- Formalized **temporal knowledge leakage** in LLM backtesting: models exploit post-cutoff pre-training knowledge when evaluated on historical prediction tasks (legal case outcomes, salary estimation, stock ranking), inflating reported accuracy and invalidating evaluation.
- Built the statistical foundation of the program: proved the leakage inflation is *non-identifiable* from passive backtest scores (entangled with skill and recency), and showed one external reference (a clean control model or the training-cutoff discontinuity) restores identification—yielding a **leakage-adjusted score** with confidence intervals.

- Introduced **Shapley-DCLR**, a claim-level metric that decomposes prediction rationales into atomic claims and applies Shapley values to quantify what fraction of *decision-critical* reasoning is contaminated—making contamination measurable and interpretable.
- Designed **TimeSPEC**, an inference-time architecture that interleaves temporally filtered retrieval with claim-level supervision, grounding predictions entirely in pre-cutoff evidence without any retraining; ablations across three LLMs show retrieval and supervision are jointly necessary.
- Developed **TEMPO**, a GRPO-based post-training method with a two-mode reward that drives leaked claims to zero as a hard prerequisite before optimizing task performance; proved the training procedure monotonically decreases leakage and converges to the leak-free optimum.
- Reduced leakage from 2–13% to 0.6–3.7% across three prediction tasks and two models, while *improving* task performance by 6–13% where strong pre-cutoff signals exist.

LAMP: Black-Box Interpretability via Locally Linear Decision Surfaces — **Northwestern University**
Jan 2024 – Jan 2025
Advisor: Prof. Bradly C. Stadie → paper [C1], AISTATS 2026 Spotlight

- Co-developed the **Linear Attribution Mapping Probe (LAMP)**, which treats an LLM's self-reported explanations as a coordinate system and fits locally linear surrogates to test whether stated factors actually steer the model's predictions.
- Designed perturbation-based probing that extracts decision surfaces *without* gradients, logits, or internal activations—enabling practical auditing of proprietary LLMs.
- Ran experiments across sentiment analysis, controversial-topic detection, and safety-prompt auditing; extracted surfaces align with human judgments and, on a clinical case-file dataset, with expert assessments.

Factual Memorization and Learning in Large Language Models — **Northwestern University**
Jan 2024 – Jan 2025
Advisor: Prof. Bradly C. Stadie (Qualifying Exam research)

- Analyzed LLM memorization under SFT and DPO fine-tuning, reproducing and extending Stanford's FinetuneBench; evaluated robustness to rephrasings and temporal shifts.
- Proposed a formal distinction between **passive memorization** (exposure) and **positive memorization** (direct QA supervision), showing the latter is markedly more efficient.
- Showed injected future-dated facts do not generalize beyond training, and that a lightweight system prompt enforcing temporal consistency mitigates the resulting overfitting.

CUOLR: Unified Off-Policy Learning to Rank — **Princeton University (remote)**
May 2022 – Dec 2023
Mentors: Prof. Mengdi Wang, Prof. Huazheng Wang → paper [C2], NeurIPS 2023

- Unified ranking under general stochastic click models as a **Markov Decision Process**, enabling principled application of offline RL to off-policy learning to rank.
- Proposed **CUOLR**, a click-model-agnostic algorithm requiring no explicit debiasing or prior knowledge of the click process; instantiated with DQN, SAC, BCQ, and CQL.
- Achieved state-of-the-art performance on large-scale real-world benchmarks with consistent robustness across heterogeneous click models.

Teaching Experience

Graduate Teaching Assistant, Department of Statistics and Data Science, Northwestern University.

STAT 303-3	Data Science 3 with Python *(Prof. Emre Besler)*	*Spring 2025, Spring 2026*
STAT 303-2	Data Science 2 with Python *(Prof. Emre Besler)*	*Winter 2025, Winter 2026*
STAT 348	Applied Multivariate Analysis *(Prof. Thomas Severini)*	*Fall 2025*
STAT 210	Introduction to Probability and Statistics *(Prof. Maxim Sinitsyn)*	*Fall 2024*

Honors & Awards

- **Spotlight Presentation**, AISTATS 2026 (LAMP). 2026
- **China National Scholarship**, Ministry of Education of the PRC (top 1%). 2020
- **Outstanding Student Scholarship** (Class A, top 1%), USTC. 2020
- National Undergraduate Electronic Design Contest: 2nd Prize (national), 1st in Anhui Province. 2021
- Scholarship for Talent Program in Basic Disciplines (Class A, top 3%), USTC. 2020

Skills

- **Programming:** Python (PyTorch, Hugging Face Transformers), R, C/C++, Bash, Git, LaTeX.
- **ML / LLM:** RL post-training (GRPO, DPO, RLHF), LLM evaluation and benchmarking, retrieval-augmented pipelines, offline reinforcement learning, API-based LLM systems (OpenAI platform).